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Project Report On
Phishing URL Detection

Semester IX
DATA AND KNOWLEDGE MINING (IS-212) PROJECT

Submitted By

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My personal interest in a growing threat of phishing attacks inspired me to make this topic, aiming to explore how data and knowledge mining techniques may effectively detect malicious URLs and protect users online.

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CHAPTER (1)

INTRODUCTION

Due to the rapid expansion of web-based services, digital communications and online payment transactions, internet usage has become an important part of our daily life. However, this growth has been accompanied by a significant rise in cyber threats among which phishing remains one of the worst and most common tricks. Phishing is a form of social engineering where attackers pretend to be real companies, banks and social networks or business services to get people to give them private information like credit card numbers, login information and personal identity data.

Most phishing attacks rely on malicious URLs designed to look a lot like real websites. Therefore, we really need a smart detection system that can identify phishing URLs so that people don't fall for cyber scams.

1.1 Project Background

As a result of the rapid digitalization of global business processes, banking operations and personal interactions, website traffic has become a critical pillar of the modern economy. As reported by cybersecurity think tanks such as APWG's (Anti-Phishing Working Group) Threat Intelligence division, phishing-related incidents and attacks have reached record-breaking levels in recent years. URL-based phishing comprises the largest category of attacks, employing a wide range of social engineering techniques to target users and compromise their personal data as well as financial accounts.

In the context of phishing, URLs are used to lure visitors into malicious websites which imitate the interfaces of popular websites. In order to appear more credible, phishing domains often use visual cues and graphics similar to those used by the original websites. To prevent phishing attempts, users are encouraged to look for security measures such as HTTPS-encrypted connections however, attackers often use dynamic programming and typosquatting to generate domains which are challenging to differentiate from the original. Additionally, phishing URLs may use subdomains hosted on free platforms or use domains generated with typo variations of the target URL (ex: g00gle.com instead of google.com).

Data and knowledge mining can solve this problem by building a proactive and automated defense mechanism. By analyzing patterns in URLs rather than a fixed set of known phishing URLs, data and knowledge mining allows the creation of flexible prediction models that can identify new phishing attempts by protecting end users before harm occurs.

1.2 Objectives of the project

The main objectives of this project are:

- To detect phishing URLs using data mining methods on the basis of selected URL structural, lexical and domain-based features.
- To conduct URL analysis in terms of features, patterns and irregularities that can be used to differentiate phishing URLs from legitimate ones.
- To specifically focus on features relevant to real time URL detection for the purpose of discovering zero-day phishing information.
- To ensure that detection results are highly accurate and can be easily interpreted by both analysts and users.

1.3 Overview / Scope of the project

This project focuses on building a data mining solution which would be able to identify phishing URLs in order to avoid users from disclosing their private information. The data mining system will extract characteristic patterns from URLs in order to classify them as either legitimate or phishing ones. The main scope of the project includes:

- **Data Analysis:** Making use of a public dataset of labeled legitimate and phishing web links.
- **Feature Extraction:** Focuses strictly on URL-level patterns (special characters and domain structures) that can be extracted quickly without visiting the actual web page.
- **Model Building:** Applies and compares standard data mining models to accurately predict phishing attempts.
- **Insight Generation:** Recognizing significant anomalies and patterns in the structure of the URL domain and risk factors.

1.4 Overview of the Methodology

The methodology of this project concentrates on suggesting a data mining approach to analyze and classify web links in order to recognize phishing URLs. The main steps are as follows:

- **Data Collection:** Data related with web links including URL string, domain properties and security labels are collected from benchmark phishing datasets thus forming a reliable and compressive dataset.
- **Data Preprocessing:** To prepare data for data mining, common preprocessing steps such as handling missing values, extracting structural URL features and normalizing features are carried out in this step.
- **Data Mining and Classification:** Predicting phishing URLs are performed using machine learning models such as HistGradient classifier, Random Forest, Linear Regression, and XGBoost.
- **Model Evaluation:** The trained models are assessed using standard performance metrics such as accuracy, precision, recall, F1-score to ensure reliable predictions in this step.
- **Pattern Analysis:** Association rule mining discovers frequent patterns in relation to URL structure, domain type and special characters.

This methodology provides a scalable, automated and interpretable approach to detect malicious URLs and support informed cybersecurity decision-making.

1.5 Benefit of the project

This project has a variety of different benefits for both the cybersecurity field and the question of threat detection:

- **Automated Classification:** By utilizing data mining and machine learning, the project allows for automated classification of web link safety which will reduce time spent checking links manually.
- **Scalable Monitoring:** The project can be used on large data sets and can easily update by utilizing new data which will allow for ongoing monitoring of phishing trends and online threats.
- **Support for non-technical audience:** By simplifying the analysis of URL features, the project has a benefit of being able to allow non-technical stakeholders to understand threat intelligence and make informed decisions about safe browsing
- **Identification of Key Risk Factors:** The project can provide what the main structural drivers of fake web links are, giving crucial information on what features should be focused on when detecting phishing URLs as well as the URLs that should be examined.
- **Contribution to Cybersecurity Defenses:** The project can benefit the cyber defenses of users and organizations by providing a computational framework which uses predictive modeling and provides threat intelligence.

This project provides a practical and interpretable framework which assists with detecting phishing URLs through the combination of predictive modeling and threat intelligence.

CHAPTER (2)

DATA PREPARATION AND PREPROCESSING

Data preparation and preprocessing are essential stages in this project. Freely available data cannot always be used without verification due to the possibility of mistakes in it. Preprocessing is also important because using unnecessary features or unclean data can lead to incorrect statistics and misleading results. Proper preparation ensures that the dataset is suitable for data mining, classification and pattern analysis.

2.1 Dataset and Data Statistics

The dataset for this project is obtained from well-known cybersecurity repositories from Kaggle including open-source URL databases. It provides detailed information on web links including special symbol counts, domain structures, protocol details, subdomains and binary target labels indicating whether a URL is legitimate or phishing. However, not all available attributes are used in the current project. This dataset is open, freely available and considered a reliable source.

The dataset contains a total of 9,944 web links collected to capture diverse structural variations across both safe and malicious sources. The distribution of URLs across different target categories is as follows:

- **Legitimate URLs (0):** 6,491 URLs (65.28%)
- **Phishing URLs (1):** 3,453 URLs (34.72%)

2.2 Data Preprocessing

Data preprocessing is an important step to ensure that the dataset is clean, consistent and suitable for data mining. Since the dataset contains lexical URL attributes, special character features and some missing or redundant values, necessary preprocessing techniques are applied before modeling.

2.2.1 Handling Missing Values

In the raw data set, missing observations are explicitly encoded as placeholders in the data frame (i.e., -1). It is essential to preprocess these observations since there is a possibility of obtaining biased estimates when training algorithms.

Missing values in directories and parameters accounts for over 50% of the entries in the data set. The features are not critical to the classification model because the primary URL features and domain-level features exist. The high missingness columns are not required in model training as their imputation can introduce unwanted bias. Thus, 56 columns with over 50% of missingness are discarded to train a robust model.

Missing values in critical continuous and categorical attributes are addressed by replacing them with an appropriate statistical estimate to avoid biased results during model training.

Effective preprocessing of missing values in the data set prevents the reporting of inaccurate results and improves the reliability of classification models.

2.2.2 Data Transformation

While extracting features such as lexical URL properties and network response indicators, individual records contain values in various numerical scales and measurement units. When feeding such data directly into machine learning algorithms, features with higher magnitude (e.g. `time_domain_activation` or `asn_ip`) could overpower the algorithm's optimizations and weight the results unreasonably. To maintain feature fairness and computational consistency, standard feature scaling techniques are applied.

As many features within the provided set represent numerical measures and discrete counts (e.g. URL length, special symbol frequencies and TTL values), feature standardization is applied before fitting the classification models. Through standard scaling, every column in a predictor matrix is adjusted to take on a mean value of 0.0 and standard deviation of 1.0, thereby transforming all numerical measures into standardized feature vectors for fitting the binary classification models.

The target classification label (phishing) is held separate from the predictor matrix before fitting the classification models. By this, the 6,491 legitimate and 3,453 phishing URLs are kept as binary ground truth labels for evaluating the outcomes. After this transformation, the features set maintains all 55 input variables to fit the classification models while standardizing all continuous and discrete measures of all 9,944 records without breaking any existing patterns.

This transformation allows utilizing different URL and domain-level attributes in training the classification models and other data mining methods such as association mining.

2.2.3 Feature Selection

Feature selection is performed to remove features that may are not useful for training of the model or cause issues such as bias, overfitting or leakage in the classification process.

Features that pose little to no beneficial information for the outcome of the model should be removed. For instance, constant features with zero variance (constancy = 1) are ignored during the training process. There are 15 constant attributes including qty_hashtag_url, qty_slash_domain, qty_questionmark_domain and qty_equal_domain in this study. These attributes contain no informative or predicting values for the phishing URL detection model and are therefore removed.

The features qty_slash_url ($r=0.6929$), length_url ($r=0.4819$) and time_domain_activation ($r=-0.4771$) have a high correlation with the target variable, URL legitimacy. Therefore, attributes that have a high correlation with the target variable should not be removed. Irrelevant or weaker attributes may lead to overfitting. In an overfitted model, instances are likely to be predicted correctly in a training set, but they have less chance of being predicted correctly in the test set. By eliminating weaker predictors, this phishing URL classifier can prevent overfitting and focus on key predictors of URL legitimacy and threats.

Finally, features with the lowest absolute value correlation coefficient (i.e. 0.05) should be cut because they provide non-informative information about the predicted variable. A total of 24 features including qty_slash_url, length_url, time_domain_activation_domain_spf, asn_ip, qty_redirects, url_shortened and others were selected and narrowed down to the most relevant predictors. The final shape of the dataset reduces from (9,944, 56) to (9,944, 25).

By using the feature selection method to narrow down the number of attributes for this project, this model will have a lower chance of being biased, overfit or leaking data. This model will be generalized and have higher reliability in making predictions and practical interpretations of URL legitimacy and threats.

2.2.4 Data Normalization

Data normalization guarantees that the information was represented consistently during the model training procedure. Since the identified feature set contains attributes that demand Min-Max normalization due to the different measure scales and value ranges including length_url, time_domain_activation and certain symbols quantity, the current set of 24 selected predictors was normalized. It implies that all attributes including the remaining continuous and discrete variables were scaled to a unit interval from 0.0 to 1.0. The corresponding Min-Max scaling transformation is applied to the 24 selected attributes to ensure that the lowest value is represented with 0.0 and the highest value is represented with 1.0. For instance, qty_dot_url was found to have -0.969730 as the lowest and 31.728844 as the highest value.

Tree-based algorithms including Random Forest and XGBoost are insensitive to the input features' scale and thus no transformation is required for such algorithms as they fit data distributions. Target label phishing was not normalized to keep the same distribution of negative and positive classes for both train and test sets. For example, there were 6491 legitimate instances and 3453 phishing instances in the given data set of 9944 records. As a result, after normalization, there are no missing values in the transformed feature set of 9944 instances and 25 attributes.

The choice of this approach also ensures that all features contribute to the final model outcome equally. In addition, it can be assumed that the URL structure and domain-based attributes are critical for classification or pattern recognition algorithms in the given task.

2.3 Data Visualization

Data visualization can be used to explore patterns and distributions in the data set. This helps to understand the characteristics and security metrics of the URL. The visual analysis is useful to recognize dominant feature correlations, target class distributions and unusual patterns that might affect the choice of features and model design.

2.3.1 Website Class Distribution

The distribution of websites across binary classification categories illustrates the relative abundance of legitimate and phishing instances within the dataset.

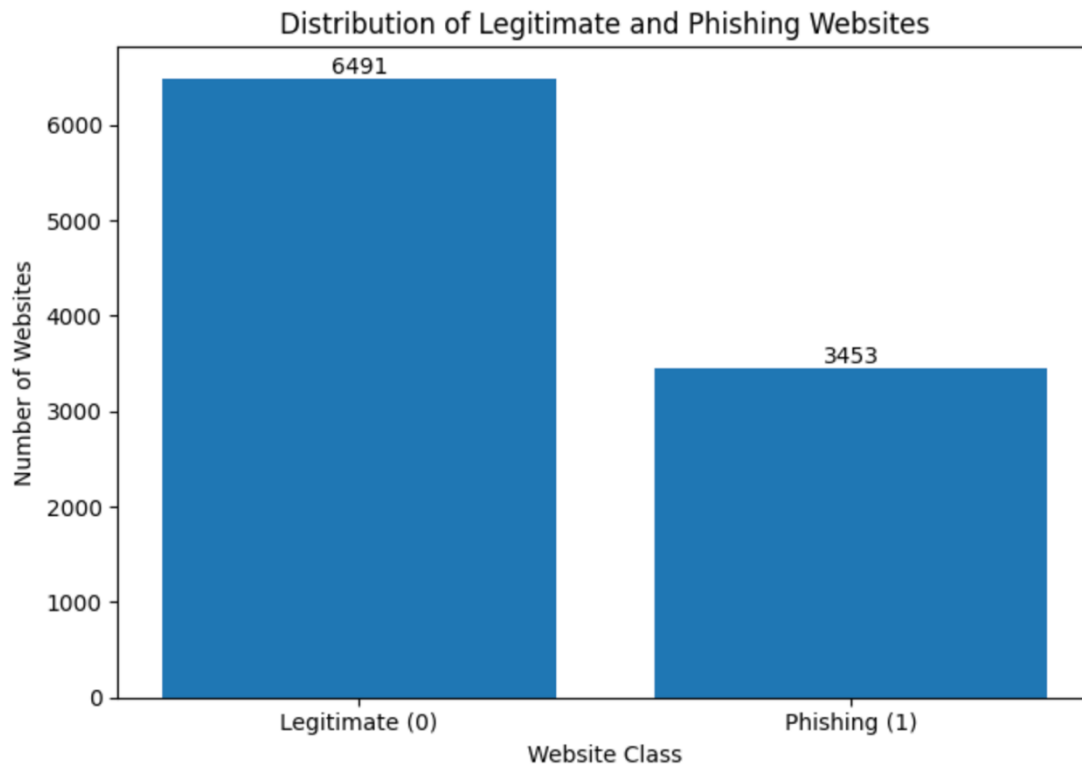


Figure 2. 1: Distribution of Legitimate and Phishing Websites

As shown in Figure 2.1, the majority of instances are classified as Legitimate (6,491 samples), while Phishing websites account for a smaller proportion of the dataset (3,453 samples).

2.3.2 Top Feature Correlation Analysis

The correlation analysis between input attributes and the target label illustrates the relative predictive strength and directional influence of key lexical and domain-level indicators.

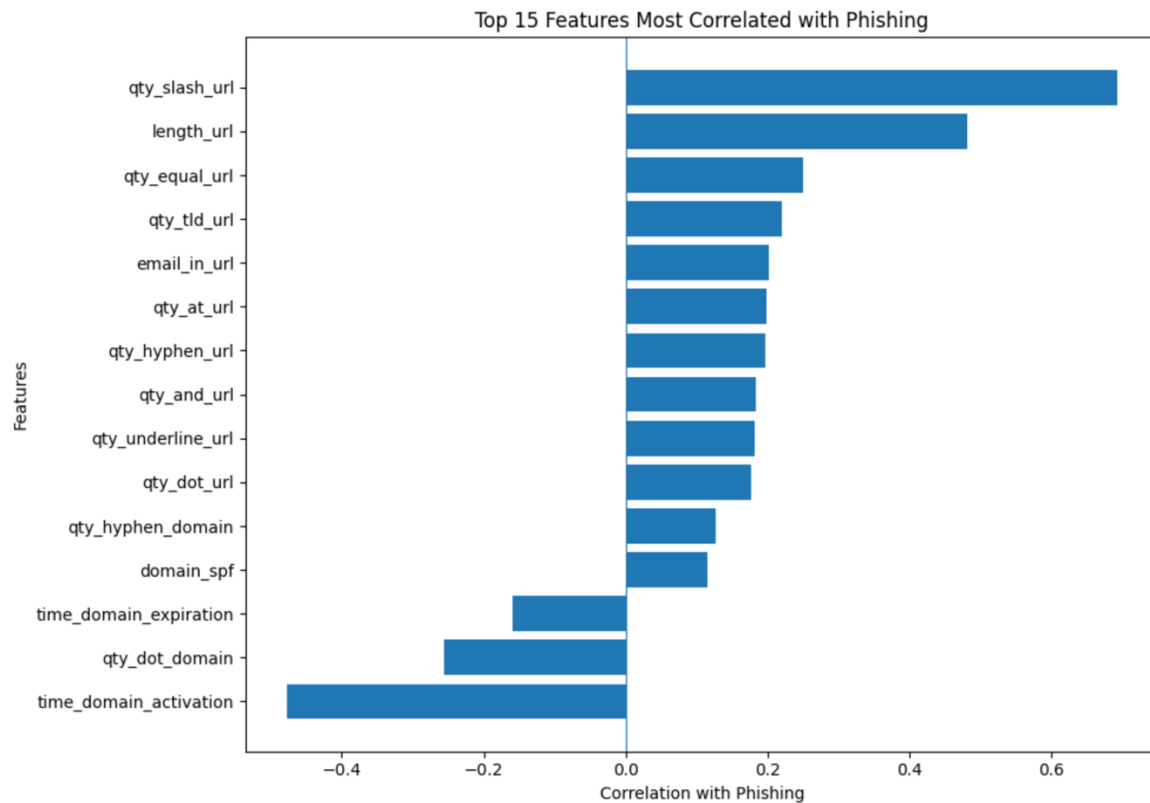


Figure 2. 2: Top 15 Features Most Correlated with Phishing

As shown in Figure 2.2, features such as `qty_slash_url` and `length_url` exhibit strong positive correlations with phishing, whereas operational metrics like `time_domain_activation` demonstrate strong negative correlations.

2.3.3 Correlation Heatmap Analysis

A correlation heatmap is generated to evaluate inter-feature relationships and potential multicollinearity among the top selected attributes.

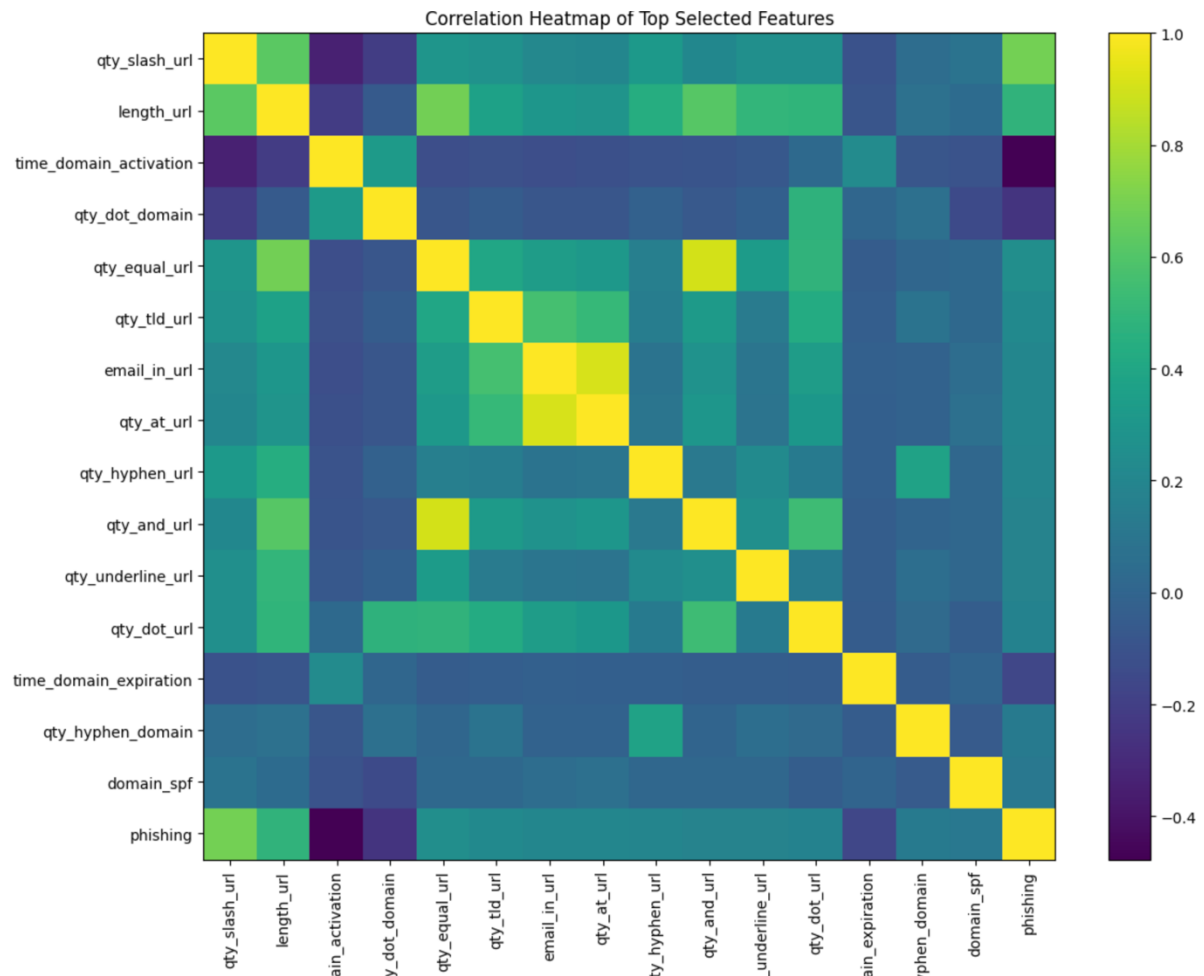


Figure 2. 3: Correlation Heatmap of Top Selected Features

As shown in Figure 2.3, features such as `email_in_url` and `qty_at_url` exhibit strong mutual correlation, while `qty_slash_url` shows high positive correlation directly with the target variable `phishing`. Evaluating these pairwise relationships ensures feature independence and helps prevent redundancy in subsequent model training.

CHAPTER (3)

APPLIED METHODOLOGIES AND MINING

Data mining techniques and machine learning are used to analyze the factors that affect the structural and domain-related aspects of the website legitimacy to build a predictive model that classifies websites based on the characteristics of phishing URLs. Both descriptive and predictive mining approaches are implemented to provide comprehensive analytical insights.

3.1 Descriptive Mining

Descriptive mining is used to discover the patterns and associations in the data that may help understand the structures of the set without specifically targeting optimization of a particular outcome. For this project the descriptive analysis is used to find out features that are somehow structurally or behaviorally linked with malicious or phishing web URLs. The process involves association rule mining and feature discretization.

3.1.1 Association Rule Mining

Association rule mining was used to find hidden dependencies and combinations of characteristics that could differentiate phishing URLs from legitimate ones. The continuous variables were discretized taking specific ranges for each category (low, medium, high). With that, the binary transaction matrix was formed which consisted of 15 attributes characterizing phishing URLs.

Feature transactions are evaluated using the Apriori algorithm to identify frequent itemsets across 9,944 instances. Rule generation produces 6,040 association rules prioritized by support, confidence and lift metrics to capture high-probability feature relationships.

The mining process identifies three core association patterns within the URL dataset:

- **Phishing Behavior Patterns:** Identifies the suspicious features discovered by the model during the analysis of phishing websites. For example, a high number of slashes in the URL ($qty_slash_url = High$) and the presence of particular domain authentication records ($domain_spf = High$) are signs of phishing. This combination means that attackers use long addresses with many additional symbols and correctly set up domain SPF records.

- **Legitimate Website Patterns:** Highlights patterns present within safe and normal websites. For example, websites with a link having a low number of equals signs ($\text{qty_equal_url} = \text{Low}$) are highly likely to have a low number of ampersand symbols ($\text{qty_and_url} = \text{Low}$) and be a member of a legitimate website category. This is due to the fact that such normal websites usually have links with less complex patterns than dangerous URLs.
- **Feature-to-Feature Relationships:** Reveal natural relations of link structures regardless of the presence of secure or unsafe sites. For example, URLs that have the $\text{qty_ip_resolved} = \text{Medium}$ are linked with URLs that have $\text{qty_slash_url} = \text{Low}$. This association rule is strict for $\text{qty_and_url} = \text{Medium}$.

By transforming these data patterns into tangible feature relationships, it was found that lexical plus domain properties are not independent but instead from distinct patterns which contribute to differentiating between legitimate and fraudulent websites.

3.2 Modeling

A modeling process is performed to classify website legitimacy on the basis of lexical, structural and network characteristics. In contrast to descriptive mining, predictive mining is geared toward discovering associations between input variables and an outcome of interest and hence involves a process of prediction.

3.2.1 Applied Methodologies / Algorithms

In this study, four supervised machine learning algorithms are applied to classify websites URLs according to their legitimacy using lexical, structural and network features. All algorithms are implemented in a manner that is consistent with the data set and the project goals.

Logistic Regression

Logistic Regression was used to predict the likelihood of URLs belonging to a legitimate or phishing category based on the weights of the structural features. It is a simple linear classifier which is used as a reference for comparison to more complex algorithms.

Random Forest

Random Forest is an ensemble method that combines bagging and decision trees. It trains a set of decision trees on random subsets of the data and aggregates their results. The algorithm allows capturing non-linear relationships between URL attributes and classes and reducing overfitting.

HistGradientBoosting (Histogram-Based Gradient Boosting)

HistGradientBoosting discretizes continuous numerical features into integer bins, which speeds up the training process tremendously. For tabular data with many features and/or skewed distributions, it can be a good choice, since it also has a small memory footprint.

XGBoost (Extreme Gradient Boosting)

XGBoost is based on gradient boosting, a technique which trains decision trees sequentially correcting the errors of its predecessors. It is efficient for capturing complex interactions between features, optimizing gradient loss functions and improving the classification of structured URL security problems.

The application of all these algorithms to the given dataset thus allows for thoroughly comparing different predictive models and choosing the one that is most efficient at identifying phishing URLs.

3.2.2 Model Set up

Model setup uses preprocessed data to train models based on machine learning to classify the legitimacy of websites.

Data Splitting

The preprocessed dataset was split into a training set (80%) and a test set (20%) using stratified sampling to ensure that both subsets maintained similar proportions of classes as the original data. The data was further split using a random state of 42 to ensure consistency in partitions during all subsequent analyses.

Target Labels

The target variable, phishing, is binary encoded, taking values 0 or 1 where 0 indicates a legitimate URL and 1 indicates a phishing URL. The extracted lexical, structural and domain features were evaluated against the targets.

Feature Extraction

The features were divided into four categories: the number of characters in the URL, the number of characters in the domain, domain-related features and security features. For continuous variables StandardScaler was applied during which the standardization parameters were fitted only on the training set which was then transformed and applied to the test set to obtain a standardized real-world input vector. Features that showed zero variance (i.e. unique value across all samples) were removed to reduce the number of dimensions.

Model training

Multiple classification algorithms were trained on the preprocessed feature vectors including the random forest, logistic regression, hist gradient boosting and XGBoost. The random forest, for instance, used the default parameters `n_estimators=100` and `n_jobs=-1`, for ensemble learning. Furthermore, the study also designed an inference pipeline that constantly extracted character distribution, domain-level properties, presence of IP address, SSL and shortening services from the input URL to produce the final phishing probability of the link using the trained classifiers.

This setup provides a structured, scalable framework for training, evaluating and deploying models to classify phishing URLs accurately.

3.2.3 Baseline Model

A baseline model was established using Logistic Regression which serves as a reference for comparison with more elaborate algorithms. Logistic Regression is used to predict the likelihood of a URL being legitimate/phishing based on lexical, structural and network characteristics of the domain.

The baseline model acts as a simple descriptive model against which the performance of more complex algorithms such as Random Forest, HistGradientBoosting and XGBoost can be compared thus enabling an assessment of the added value of each subsequent algorithm in terms of increased predictive accuracy.

3.2.4 Proposed Model and Data Insights

The proposed model for this project is XGBoost. The reason for selecting this model is its capability to capture nonlinear dependencies between the lexical, structural and semantic features of the URL and effectively handle high-dimensional structured data sets of URLs. This algorithm is efficient since it constructs decision trees sequentially to minimize gradient errors during the iterative process leading to accurate predictions of phishing URLs by lowering FN.

XGBoost algorithm was implemented to classify phishing URLs and assess the importance of each feature for the model. The analysis revealed that structural and lexical features were the most informative for training the classifier. For instance, the maximum likelihood of URL classification depends on the total length of the URL, the length of the domain name, special characters such as dots, hyphens and slashes and the HTTPS status.

Therefore, the XGBoost algorithm is an appropriate method to address the problem of phishing URL detection because of its advantages of achieving performance optimization when classifying URLs and identifying the key features for training the model.

CHAPTER (4)

EVALUATION

A logistic regression, random forest, hist gradient boosting, and XGBoost machine learning classifiers were trained on the same set of features to evaluate the accuracy of URL legitimacy prediction. The results of the classification were assessed using accuracy scores, stability and consistency.

4.1 Performance Metrics

The models were evaluated using binary classification metrics to assess the predictive performance on the URL categories. The evaluation includes Accuracy, Precision, Recall and F1 score. To provide a balanced assessment of class imbalance cases, both macro and weighted averages were reported.

Macro averages are the mean values of the metrics for the two classes, which treat both classes equal. Weighted averages consider the distribution of samples between the classes. In addition, the evaluation used confusion matrices which display the class distribution at the prediction stage.

4.2 Comparison of Experiment Results/Model Comparison

The predictive performance of the four models was compared using the macro and weighted averages of precision, recall, and F1-score. The results are summarized in the table below:

Model	Macro Precision	Macro Recall	Macro F1-Score	Weighted Precision	Weighted Recall	Weighted F1-Score	Overall Accuracy
XGBoost	0.92	0.92	0.92	0.93	0.93	0.93	92.96%
HistGradientBoosting	0.93	0.92	0.92	0.93	0.93	0.93	93.16%
Random Forest	0.92	0.92	0.92	0.93	0.93	0.93	92.86%
Logistic Regression	0.92	0.90	0.91	0.92	0.92	0.92	91.70%

Table 4. 1: Performance Comparison of Classification Models

Model	TP	TN	FP	FN
XGBoost	622	1227	71	69
HistGradientBoosting	612	1241	57	79
Random Forest	612	1235	63	79
Logistic Regression	583	1241	57	108

Table 4. 2: Confusion Matrix across Models

Based on the given comparison, it can be concluded that the application of the ensemble method significantly improved the results compared to the baseline model of Logistic Regression.

In the case of phishing detection, minimizing the number of False Negatives (FN) failing to identify an actual phishing URL is more critical than overall accuracy as undetected malicious sites put the users at risk of being compromised directly.

The confusion matrix analysis indicated that XGBoost achieved a higher Phishing Recall (0.90 / 90%) and successfully captured 622 true phishing samples resulting in only 69 false negatives compared to HistGradientBoosting with 79 false negatives. As a result, XGBoost was selected as the proposed primary classification model for the real-time URL evaluation.

4.2.1 Evaluation Analysis

The evaluation results confirm the superiority of ensemble-based solutions over the baseline Logistic Regression model. While HistGradientBoosting algorithm demonstrates the highest numerical accuracy and weighted F1-score on the static test split, XGBoost displays a strong overall performance due to effective feature representation and variance reduction via boosting.

The analysis of the confusion matrix reveals that the majority of the misclassified entries are caused by phishing URL's similarities with the legitimate web addresses, which highlights subtle lexical and domain specific patterns. Additionally, the minority class imbalance affects the macro average as opposed to the weighted performance metric.

While HistGradientBoosting demonstrates slightly higher accuracy and F1-score on the test set, XGBoost possesses a better generalization capability reflected in higher phishing recall and lower false discovery rate for the real-world application.

4.2.2 ROC Curve and AUC

The predictive accuracy of XGBoost was further evaluated using the ROC and AUC metrics.

The AUC values help understand how well the model can distinguish the legitimate web targets from malicious phishing URLs in terms of different classification thresholds.

Accordingly, it is revealed that XGBoost achieves a considerably high AUC of 0.9779 implying that the algorithm can efficiently differentiate legitimate and phishing URL classes thus successfully addressing the threat.

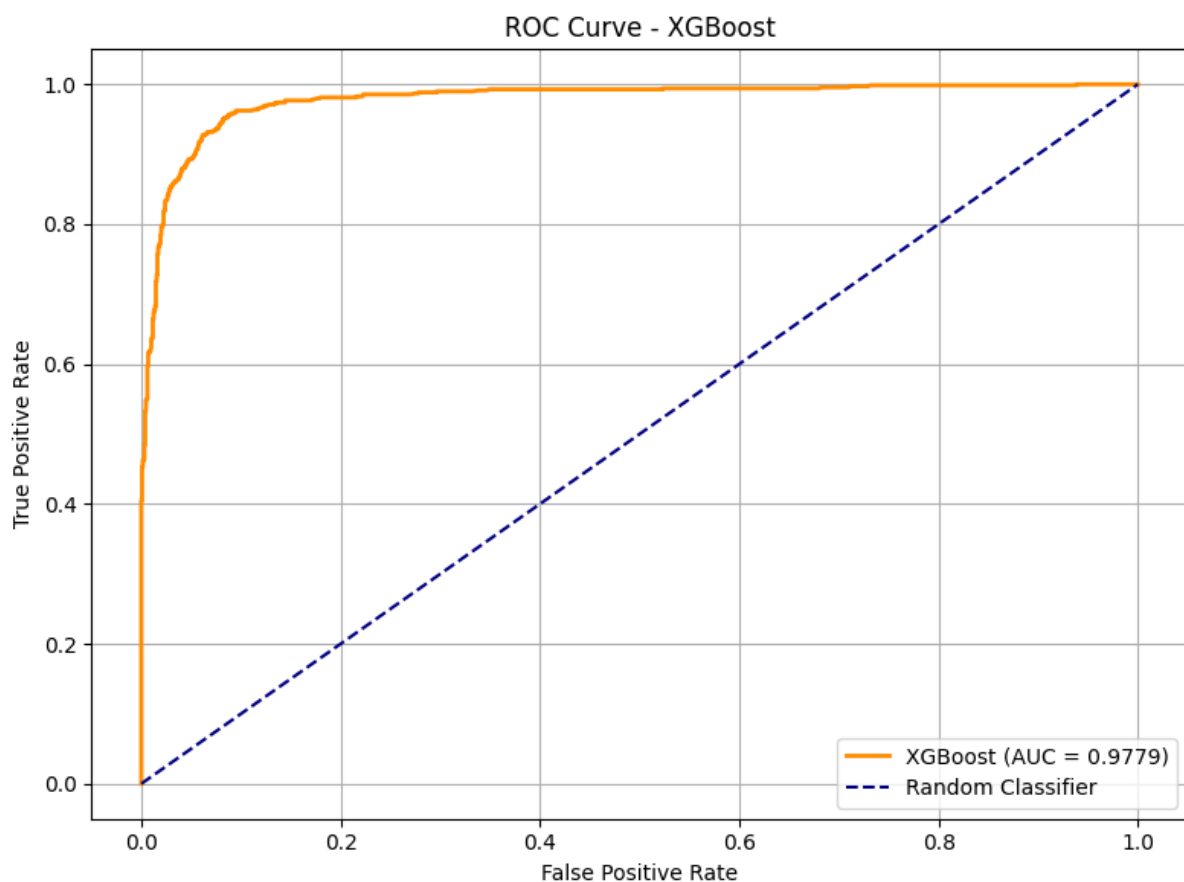


Fig 4. 1: ROC Curve of XGBoost

4.2.3 Cross Validation

To evaluate model stability and generalization, 5-fold stratified cross-validation was performed for all models. Each model was trained on four folds and validated on the remaining fold repeating this process five times. The mean accuracy and standard deviation across folds are reported in the following table.

Model	Mean Accuracy	Stability (Standard Deviation)
XGBoost	92.55%	$\pm 0.46\%$
HistGradientBoosting	93.05%	$\pm 0.53\%$
Random Forest	92.74%	$\pm 0.38\%$
Logistic Regression	90.01%	$\pm 0.15\%$

Table 4. 3: Cross-Validation Accuracy and Stability for All Models

The results indicate that HistGradientBoosting achieves the highest mean accuracy while maintaining low variability across folds, demonstrating both strong predictive performance and stable generalization. Random Forest and XGBoost also show competitive accuracy through XGBoost exhibits slightly higher variability, while Logistic Regression performs slightly lower confirming its role as a reliable baseline.

4.2.4 Findings

While HistGradientBoosting showcase exceptional numeric performance on the static benchmarks (93.05% $\pm 0.53\%$ accuracy for cross-validation), XGBoost is the most practical classifier when it comes to real-time phishing URL detection. The model with high boosting accuracy also demonstrates competitive performance while Logistic Regression plays the role of a simple baseline algorithm.

The analysis of the confusion matrix and the metric trends reveal that the majority of misclassifications come down to similarities between the legitimate and phishing URLs on the structural level. While other algorithms concentrate too much on the global performance, XGBoost manages to achieve better results for the positive class reflected in the higher phishing recall (0.90) and a lower number of false negatives (69 FN) compared to HistGradientBoosting (79 FN). Finally, the high ROC-AUC values (0.9779) across all models highlight the classifier's ability to distinguish between the two classes.

The overall evaluation demonstrates that XGBoost provides reliable accuracy, precision, and recall which makes it a good candidate for the real-time phishing URL detection system. Its performance characteristics are better in most of the considered dimensions compared to other classifiers, making it a viable option for solving the given problem.

CHAPTER (5)

CONCLUSION

The findings demonstrate that machine learning particularly XGBoost can effectively classify URL legitimacy. The system provides both prediction and threat mitigation capabilities for web security analysis.

5.1 Advantages of using this project

The project's results allow making accurate and reliable predictions of phishing attempts; the most common lexical and structural URL features are identified in the process. The model is stable and highly accurate which was confirmed by cross-validation, ROC-AUC analysis(AUC = 0.9779) and low rates of false negatives.

It helps predict phishing URLs which can be used to prevent cyber-attacks and it is possible to integrate the model into web security filters or browsers' extensions. The features identified by the project can help understand what kind of structural characteristics in URLs can make them phishing ones.

5.2 Disadvantages of using this project

The accuracy of prediction is based on the features extracted from URLs and malicious attackers can easily come up with zero-day obfuscation methods to avoid detection by lexical patterns. The accuracy of the model also depends on the quality and freshness of the data used to train the model. In addition, the analysis at the domain level deals only with structural properties and ignores other aspects of the web page's behavior such as redirections triggered by Java-scripts or modifications of the structure.

In general, the tool can be considered effective in predicting phishing URLs while also providing insights into the critical lexical features that influence the results. The lexical analysis can be applied for cybersecurity purposes to detect phishing attempts.

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